Denny Schaedig

Machine Learning Engineer

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SUMMARY

ML engineer who ships ML systems from data pipelines and distributed training to packaging and serving. Published researcher in generative models, LLM agents, rigorous evaluation, and HIPAA-compliant cloud/GPU infrastructure.

SKILLS

Languages: Python, SQL, TypeScript / JavaScript, Dart, Bash

ML & LLMs: PyTorch, TensorFlow / Keras, Hugging Face Transformers, scikit-learn; GANs, diffusion, transformers, computer vision, multimodal fusion; LLM agents, RAG, LLM evaluation

Deployment & MLOps: AWS (HIPAA HPC), Docker, Ray, CUDA, Prometheus, FastAPI, TensorFlow Lite, Hugging Face Hub, PyPI, GitHub Actions; fMRI / fNIRS data

PROJECTS

- **Maxim** (pymaxim.bio) - Open-source LLM agent harness (~13K PyPI downloads) with pre-registered, ablation-controlled evals, nulls included; a taught want transferred between agents (0.84 vs. 0.22, n = 50 per arm).

EXPERIENCE

Laboratory for Child Brain Development — Washington University in St. Louis — St. Louis, MO
NeuroImaging Engineer (Machine Learning & Software Engineering) — *Dec 2022 – Present*

- Architected a HIPAA-compliant AWS HPC environment (EC2, EFS, S3, Glacier) ~32% under projected cost and secured an additional 24% university-wide cloud discount.
- Designed and shipped **HRfunc**, an open-source Python package for signal deconvolution; published in Neurophotonics (2025), with 22 public releases and ~18K PyPI downloads.
- Trained NIRSgen, a diffusion model for biomedical time-series augmentation; transfer learning lifted downstream accuracy from 76% to 92%.
- Developing **synchronAI**, a multimodal model fusing video, audio, and time-series signals (0.72 AUC).

Rocky Mountain Digerati, LLC — Summit County, CO
Founder & ML Engineer (independent applied-ML lab) — *Nov 2025 – Present*

- Built an LLM analyst agent for a healthcare-finance contract client, including a hallucination detection and removal layer, running on open-weight models as small as 16B parameters (in client beta).
- Run a self-hosted 3-GPU Ray cluster (CUDA + Apple Silicon, Tailscale mesh) for training, evals, and serving.
- Collected a 4,040-sample snowpack dataset (40K+ Hugging Face downloads) to pre-train snowGAN; built AvAI, a Flutter check-out safety app for iOS/Android (launching fall 2026) with a Prometheus-monitored backend.

The Pain Lab — University of Colorado Boulder — Boulder, CO
Software / ML Engineer — *Jul 2020 – Feb 2023*

- Trained an RNN to predict pain from wearable-sensor and questionnaire data (79% vs. 63% baseline).

Intermountain Neuroimaging Consortium — University of Colorado Boulder — Boulder, CO
Software Engineer — *Jul 2017 – Feb 2022*

- Built ETL pipelines and automation for MRI scheduling, study enrollment, and training compliance.

SELECTED PUBLICATIONS (5 total: dennyschaedig.com/papers)

Schaedig, D., et al. *HRfunc: A tool for modeling hemodynamic response variability in fNIRS.* Neurophotonics (2025).
Graff, K., Zhang, Y., **Schaedig, D.**, et al. *Stability of the functional connectome across childhood.* NeuroImage (2026).

EDUCATION

M.S., Artificial Intelligence & Machine Learning - Colorado State University Global — 2025
B.A., Neuroscience - University of Colorado Boulder — 2019